



भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी
Indian Institute of Information Technology Guwahati

CS236 - ARTIFICIAL INTELLIGENCE LAB
ASSIGNMENT - 08

Problem Statement

Write a Python program to solve the following optimization problems using **Genetic Algorithms (GA)**:

1. **Travelling Salesman Problem (TSP)**
2. **Optimal Power Flow (OPF) Problem**

The program must implement a population-based search with specific operators for combinatorial (permutation) and continuous (real-coded) spaces.

Algorithm Specifications

1. Travelling Salesman Problem (TSP):

- i. **Representation:** Permutation-based chromosome (city 0 is fixed at start and end).
- ii. **Crossover:** Implement **Order Crossover (OX)** to maintain valid permutations.
- iii. **Mutation:** Implement **Swap Mutation** (exchange two random non-zero cities).
- iv. **Selection:** Use Tournament Selection with size $k = 3$.

2. Optimal Power Flow (OPF) Problem:

- i. **Representation:** Real-coded chromosome $[P_{g1}, P_{g2}, \dots, P_{gm}]$.
- ii. **Simulated Binary Crossover (SBX):** For parents p_1, p_2 , generate offspring c_1, c_2 using:

$$c_{1,i} = 0.5[(1 + \beta_i)p_{1,i} + (1 - \beta_i)p_{2,i}], \quad c_{2,i} = 0.5[(1 - \beta_i)p_{1,i} + (1 + \beta_i)p_{2,i}]$$

where the spread factor β_i is computed from a random $u \sim \mathcal{U}[0, 1]$ and distribution index $\eta_c = 20$ as:

$$\beta_i = \begin{cases} (2u)^{1/(\eta_c+1)} & \text{if } u \leq 0.5 \\ \left(\frac{1}{2(1-u)}\right)^{1/(\eta_c+1)} & \text{if } u > 0.5 \end{cases}$$

- iii. **Polynomial Mutation:** For a gene x , the mutated gene x' is:

$$x' = x + (x_{\max} - x_{\min}) \delta$$

where δ is computed from a random $u \sim \mathcal{U}[0, 1]$ and distribution index $\eta_m = 20$ as:

$$\delta = \begin{cases} (2u)^{1/(\eta_m+1)} - 1 & \text{if } u < 0.5 \\ 1 - (2(1-u))^{1/(\eta_m+1)} & \text{if } u \geq 0.5 \end{cases}$$

- iv. **Fitness Function:** Minimize $F = \sum_i (a_i + b_i P_{gi} + c_i P_{gi}^2) + \lambda (\text{Power Balance Violation})^2$, where $\lambda = 1000$. Enforce $P_i^{\min} \leq P_{gi} \leq P_i^{\max}$ as hard bounds during mutation.

Experimental Requirements

Constraints on Environment

To ensure the focus remains on algorithm logic and manual implementation of operators:

- Use a basic text editor (e.g., Notepad, Gedit, Vim) without AI-assisted coding or autocompletion.
- **Prohibited:** Use of GA libraries like PyGAD, DEAP, or `Scipy.optimize`. You must implement the mathematical logic for SBX and Polynomial Mutation from scratch.
- Execution and performance logging must be handled via the command line.

Data Collection & Plotting

For both problems, generate the following plots:

- **Convergence Plot:** A graph showing the Best Fitness vs. Generation number.
- **Parameter Study:** For OPF, compare the convergence speed using $\eta_c = 2$ vs. $\eta_c = 20$.

Input/Output Format

Sample Input:

```
TSP: 4 cities
0 10 15 20
10 0 35 25
15 35 0 30
20 25 30 0
OPF: 2 gens, 100MW Demand
10 0.5 0.02 20 80
15 0.4 0.015 30 120
```

Sample Output:

```
TSP Solution: Route: 0 -> 1 -> 3 -> 2 -> 0 | Distance: 80
OPF Solution: Outputs: [42.31, 57.69] | Cost: 163.47
```

Submission Requirements

- **Source Code:** A single Python file implementing both GA variants manually.